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Electronic Principles and Circuits

BEC303 — MODULE 4

Negative Feedback & Active Filters

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PART A: NEGATIVE FEEDBACK

1. Introduction to Negative Feedback

A negative feedback amplifier is an electronic amplifier in which a portion of the output signal (voltage or current) is fed back to the input in a phase-opposite manner, opposing the original input. This reduces the net input and hence reduces overall gain — but delivers tremendous improvements in stability, bandwidth, distortion and impedance.

Benefits of Negative Feedback

- Gain Reduction with Stability: Closed-loop gain becomes stable and predictable (depends only on resistors, not A_{VOL}).
- Bandwidth Improvement: $BW_{CL} = BW_{OL} \times (1 + A_{VOL} \times B)$. Reducing gain increases bandwidth.
- Reduced Distortion & Noise: $THD_{CL} = THD_{OL} / (1 + A_{VOL} \times B)$. Distortion decreases by loop gain factor.
- Improved Input Impedance (VCVS/VCIS): Increases by factor $(1 + A_{VOL} \times B)$.
- Reduced Output Impedance (VCVS/ICVS): Decreases by factor $(1 + A_{VOL} \times B)$.
- Increased Output Impedance (VCIS/ICIS): Increases by factor — acts like ideal current source.

KEY PRINCIPLE

The price of negative feedback is reduced gain. Since op-amps provide $A_{VOL} \sim 100,000$ to 1,000,000, the trade-off is always beneficial — transforming an unreliable high-gain amplifier into a precise, stable system whose gain depends almost entirely on passive resistors.

The fundamental concept is the feedback fraction B (feedback attenuation factor) — the fraction of output signal returned to the input:

$$B = v_2 / v_{out} \quad \text{Loop Gain} = A_{VOL} \times B \quad (\text{Must be } \gg 1 \text{ for good NFB})$$

2. Four Types of Negative Feedback

Since input and output can each be either voltage or current, there are four types of negative feedback. The original Table 19-1 from the textbook is shown below:

Table 19-1: Ideal Negative Feedback — Four Types

Figure 19-1 below shows the circuit representations of VCVS and ICVS — note the HIGH z_{in} for VCVS and LOW z_{in} for ICVS:

Fig 19-1: (a) VCVS — Voltage-controlled voltage source; (b) ICVS — Current-controlled voltage source

Figure 19-2 shows VCIS and ICIS — note the HIGH z_{out} (current source symbol) for both:

Fig 19-2: (a) VCIS — Voltage-controlled current source; (b) ICIS — Current-controlled current source

Type	z_{in} (Ideal)	z_{out} (Ideal)	Ratio & Symbol
VCVS	∞ (infinite)	0 (zero)	$A_v = v_{out}/v_{in}$
ICVS	0 (zero)	0 (zero)	$r_m = v_{out}/i_{in} (\Omega)$
VCIS	∞ (infinite)	∞ (infinite)	$g_m = i_{out}/v_{in} (S)$
ICIS	0 (zero)	∞ (infinite)	$A_i = i_{out}/i_{in}$

2.1 VCVS — Voltage Controlled Voltage Source

The VCVS is a voltage amplifier where an input voltage controls an output voltage. It has stabilized voltage gain, infinite input impedance, and zero output impedance (ideal). This is the classic non-inverting op-amp configuration. In practice, input impedance is very high (M Ω range) and output impedance very low (m Ω range).

Input/Output: Voltage $\rightarrow$ Voltage

Ideal z_{in} : ∞ (infinite) — draws no current from source

Ideal z_{out} : 0 (zero) — acts as stiff voltage source to any load

Gain Symbol: $A_v = v_{out} / v_{in}$ (dimensionless)

Also Known As: Voltage Amplifier

VCVS Application

High z_{in} $\times$ Low z_{out} $\times$ Stable voltage gain. Used in: buffer amplifiers, precision voltage amplifiers, instrumentation amps, audio pre-amps.

2.2 ICVS — Current Controlled Voltage Source

The ICVS (Transresistance Amplifier / Current-to-Voltage Converter) has an input current controlling an output voltage. Implemented by connecting input current to the inverting terminal of an op-amp with feedback resistor R_f . Both input and output impedances are driven toward zero by heavy feedback.

Input/Output: Current $\rightarrow$ Voltage

Ideal z_{in} : 0 (zero) — virtual ground absorbs all input current

Ideal z_{out} : 0 (zero) — stiff voltage source

Gain Symbol: $r_m = v_{out} / i_{in}$ (units: Ohms, Ω)

Also Known As: Transresistance Amplifier, I-to-V Converter

ICVS Application

Low z_{in} $\times$ Low z_{out} $\times$ Converts current to voltage. Used in: photodetector interfaces, photodiode amplifiers, current sensors.

2.3 VCIS — Voltage Controlled Current Source

The VCIS (Transconductance Amplifier / Voltage-to-Current Converter) has an input voltage controlling a precisely stabilized output current. The load R_L is placed in the feedback path, making it both load and feedback element. Very high input AND output impedance — ideal current source behavior.

Input/Output: Voltage $\rightarrow$ Current

Ideal z_{in} : ∞ (infinite) — same as VCVS

Ideal z_{out} : ∞ (infinite) — acts as ideal current source

Gain Symbol: $g_m = i_{out} / v_{in}$ (units: Siemens, S or mhos)

Simplified Equation: $i_{out} = v_{in} / R_1$, so $g_m = 1/R_1$

Also Known As: Transconductance Amplifier, V-to-I Converter

VCIS Application

High z_{in} $\times$ High z_{out} $\times$ Converts voltage to current. Used in: constant current sources, LED drivers, current-output DACs, motor speed control.

2.4 ICIS — Current Controlled Current Source

The ICIS (Current Amplifier) amplifies input current to produce larger stabilized output current. Very low input impedance (absorbs all input current) and very high output impedance (ideal current source). Implemented using specific topology with R_1 and R_2 setting current gain.

Input/Output: Current $\rightarrow$ Current

Ideal z_{in} : 0 (zero) — low impedance to absorb all input current

Ideal z_{out} : ∞ (infinite) — ideal current source

Gain Symbol: $A_i = i_{out} / i_{in}$ (dimensionless). $A_i \approx R_2/R_1 + 1$

ICIS Application

Low z_{in} $\times$ High z_{out} $\times$ Current gain $A_i = R_2/R_1 + 1$. Used in: current mirrors, precision current scaling, current amplifiers.

3. VCVS Voltage Gain — Closed Loop Analysis

Figure 19-3 (below) shows the VCVS non-inverting amplifier. The op-amp has A_{VOL} typically 100,000+. The voltage divider (R_1 and R_f) feeds part of the output back to the inverting input.

Fig 19-3: VCVS Amplifier — Non-inverting configuration with feedback divider R_1 , R_f

3.1 Feedback Fraction and Exact Closed-Loop Gain

The feedback fraction B = fraction of output voltage fed back to inverting input:

Feedback Fraction B

$$B = v_2 / v_{out} = R_1 / (R_1 + R_f) \quad \dots (19-1)$$

Using algebraic analysis of the VCVS loop, the exact closed-loop voltage gain is:

Exact Closed-Loop Gain

$$A_v(CL) = A_{VOL} / (1 + A_{VOL} \times B) \quad \dots (19-2, 19-3)$$

3.2 Ideal Closed-Loop Gain

When $A_{VOL} \times B \gg 1$ (always true for practical op-amps), the equation simplifies dramatically:

Ideal Gain (Non-Inverting Amp)

$$A_v \approx 1/B = (R_1 + R_f)/R_1 = R_f/R_1 + 1 \quad \dots (19-4)$$

How to Calculate Non-Inverting Amp Gain

Step 1: Identify R_1 (inverting input to GND) and R_f (output to inverting input). Step 2: $B = R_1 / (R_1 + R_f)$ Step 3: $A_v \approx 1/B = R_f/R_1 + 1$ Example: $R_1 = 1\text{k}\Omega$, $R_f = 9\text{k}\Omega \rightarrow B = 0.1 \rightarrow A_v = 10$

4. Other VCVS Equations

4.1 Gain Stability

Gain stability depends on the percent error between ideal gain ($1/B$) and actual gain. Worst-case error occurs when A_{VOL} is at its minimum:

$$\% \text{ Max Error} = 100\% / (1 + A_{VOL}(\min) \times B) \quad \dots (19-7)$$

Example: $A_{VOL}(\min) = 50,000$, $B = 0.01 \rightarrow \% \text{ Error} = 100\% / (1 + 500) = 0.2\%$. Excellent stability!

4.2 Closed-Loop Input Impedance

Figure 19-5a (below) shows the VCVS with labeled $z_{in(CL)}$. Voltage-series feedback dramatically increases input impedance:

$$z_{in(CL)} = (1 + A_{VOL} \times B) \times R_{in} \parallel R_{CM} \quad \dots (19-8)$$

Since R_{CM} is very large, this simplifies to:

$$z_{in(CL)} \approx (1 + A_{VOL} \times B) \times R_{in} \quad \dots (19-9)$$

Example: $R_{in} = 2\text{M}\Omega$, $A_{VOL} \times B = 1000 \rightarrow z_{in(CL)} \approx 2\text{G}\Omega$! Extremely high.

4.3 Closed-Loop Output Impedance

$$z_{out(CL)} = R_{out} / (1 + A_{VOL} \times B) \quad \dots (19-10)$$

With $A_{VOL} \times B = 1000$ and $R_{out} = 75\Omega \rightarrow z_{out(CL)} = 0.075\Omega$. Nearly zero — drives any load.

4.4 Nonlinear Harmonic Distortion (THD)

Figure 19-5(b)(c) below shows the nonlinear distortion waveform and resulting harmonic spectrum. At 1kHz input, distortion creates harmonics at 2kHz, 3kHz, etc.:

Fig 19-5: (a) VCVS amplifier with $z_{in(CL)}$ and $z_{out(CL)}$; (b) Nonlinear base-emitter distortion; (c) Harmonic spectrum at 1, 2, 3 kHz

$$\text{THD} = (\text{Total Harmonic Voltage} / \text{Fundamental Voltage}) \times 100\% \quad \dots (19-11)$$

Closed-Loop THD

$$\text{THD}_{\text{CL}} = \text{THD}_{\text{OL}} / (1 + A_{\text{VOL}} \times B) \quad \dots (19-12)$$

With $A_{\text{VOL}} \times B = 1000$, closed-loop distortion is $1001\times$ lower than open-loop — enormous improvement for audio applications.

5. The ICVS Amplifier (Transresistance)

Figure 19-7 (below) shows the ICVS transresistance amplifier — a current-to-voltage converter. Input current i_{in} is injected at the virtual ground node (inverting terminal). The feedback resistor R_f forces all input current through it, establishing v_{out} .

Fig 19-7: ICVS Amplifier — Transresistance (current-to-voltage) configuration

The exact output voltage equation is:

$$v_{\text{out}} = -(i_{\text{in}} \times R_f \times A_{\text{VOL}}) / (1 + A_{\text{VOL}}) \quad \dots (19-13)$$

Since $A_{\text{VOL}} \gg 1$, this simplifies to the ideal equation:

Simplified: $r_m = -R_f$ (Ohms)

$$v_{\text{out}} = -(i_{\text{in}} \times R_f) \quad \dots (19-14)$$

The closed-loop impedances for ICVS are:

$$z_{\text{in}}(\text{CL}) = R_f / (1 + A_{\text{VOL}}) \rightarrow \sim 0 \quad \dots (19-15)$$

$$z_{\text{out}}(\text{CL}) = R_{\text{out}} / (1 + A_{\text{VOL}}) \rightarrow \sim 0 \quad \dots (19-16)$$

Both impedances are nearly zero — perfect for current-to-voltage conversion. Zero z_{in} means the virtual ground node is at $\sim 0\text{V}$, so all i_{in} flows through R_f .

ICVS Amplifier Summary	
Input	Current i_{in} (high-impedance source)
Output	$v_{\text{out}} = -i_{\text{in}} \times R_f$
$z_{\text{in}}(\text{CL})$	$\approx R_f / A_{\text{VOL}} \rightarrow$ nearly zero (virtual ground)
$z_{\text{out}}(\text{CL})$	$\approx R_{\text{out}} / A_{\text{VOL}} \rightarrow$ nearly zero
Transresistance	$r_m = -R_f$ (units: Ohms)
Applications	Photodiode amp, current sensor, I-to-V converter

6. The VCIS Amplifier (Transconductance)

Figure 19-10 (below) shows the VCIS transconductance amplifier. Similar to VCVS non-inverting amp, but R_L is placed in the feedback path — it is both load AND feedback element. The output monitored is the current through R_L , not voltage.

Fig 19-10: VCIS Amplifier — Transconductance (voltage-to-current) configuration

The exact output current equation is:

$$i_{out} = v_{in} / (R_1 + (R_1 + R_L)/A_{VOL}) \quad \dots (19-18)$$

Since A_{VOL} is huge, this simplifies to:

$$i_{out} = v_{in} / R_1 \quad \text{i.e. } g_m = 1/R_1 \text{ (Siemens)} \quad \dots (19-19)$$

$$z_{in}(CL) = (1 + A_{VOL} \times B) \times R_{in} \rightarrow \text{very HIGH} \quad \dots (19-20)$$

$$z_{out}(CL) = (1 + A_{VOL}) \times R_1 \rightarrow \text{very HIGH} \quad \dots (19-21)$$

With $A_{VOL} = 100,000$ and $R_1 = 1k\Omega \rightarrow z_{out}(CL) = 100M\Omega$! Acts like ideal current source.

VCIS Amplifier Summary	
Input	Voltage v_{in}
Output	Constant current $i_{out} = v_{in} / R_1$
$z_{in}(CL)$	Very high (same as VCVS)
$z_{out}(CL)$	$(1 + A_{VOL}) \times R_1 \rightarrow \text{very large (100M}\Omega\text{+)}$
Transconductance	$g_m = 1/R_1$ (Siemens)
Applications	LED driver, constant current source, motor drive

7. The ICIS Amplifier (Current Amplifier)

Figure 19-12 (below) shows the ICIS current amplifier. The ICIS circuit amplifies input current to produce a larger, stabilized output current. Heavy negative feedback gives very low z_{in} and very high z_{out} — near-ideal current amplifier behavior.

Fig 19-12: ICIS Amplifier — Current amplifier (current-controlled current source)

The closed-loop current gain:

$$A_i = A_{VOL}(R_1 + R_2) / (R_L + A_{VOL} \times R_1) \quad \dots (19-22)$$

Simplified ($A_{VOL} \times R_1 \gg R_L$):

Simplified Current Gain

$$A_i \approx R_2/R_1 + 1 \quad \dots (19-23)$$

$$z_{in}(CL) = R_2 / (1 + A_{VOL} \times B) \rightarrow \text{very small} \quad \dots (19-24)$$

$$B = R_1 / (R_1 + R_2) \quad \dots (19-25)$$

$$z_{out}(CL) = (1 + A_{VOL}) \times R_1 \rightarrow \text{very large} \quad \dots (19-26)$$

Large A_{VOL} : very small z_{in} AND very large $z_{out} \rightarrow$ almost perfect current amplifier.

ICIS Amplifier Summary

Input	Current i_{in}
Output	Amplified current $i_{out} = A_i \times i_{in}$
z_{in} (CL)	Very small: $R_2 / (1 + A_{VOL} \times B)$
z_{out} (CL)	Very large: $(1 + A_{VOL}) \times R_1$
Current Gain	$A_i \approx R_2 / R_1 + 1$
Feedback Fraction	$B = R_1 / (R_1 + R_2)$
Applications	Current mirrors, precision current scaling

PART B: ACTIVE FILTERS

8. Introduction to Active Filters

A filter is a circuit capable of passing certain frequencies while attenuating (blocking) other frequencies. Active filters use op-amps + R + C (NO inductors) and can provide voltage gain. Key advantages: eliminates bulky inductors, provides gain, high z_{in} , low z_{out} , easy to cascade.

Filter Type	Passes	Blocks	Application
Low Pass (LPF)	0 to f_c	Above f_c	Anti-aliasing, audio bass
High Pass (HPF)	Above f_c	0 to f_c	DC blocking, treble boost
Bandpass (BPF)	f_1 to f_2	Outside band	Radio tuning, equalizer
Bandstop (BSF)	Outside band	f_1 to f_2 notch	50/60Hz hum removal
All Pass (APF)	All frequencies	None	Phase shifting circuits

9. Ideal Filter Responses

9.1 Low Pass Filter

A low-pass filter passes all frequencies from DC (0Hz) up to the cutoff frequency f_c and blocks all frequencies above f_c . Ideal LPF: zero attenuation in passband, infinite attenuation in stopband, vertical (brick-wall) transition.

Fig 21-1: Ideal Low-Pass Filter Response — Passband (0 to f_c) then Stopband

- Passband: 0 to f_c — signal passes with no attenuation ($A_v = \text{constant}$)
- Stopband: above f_c — infinite attenuation ($A_v = 0$ ideal)
- Roll-off: 1st order = -20 dB/decade, 2nd order = -40 dB/decade

9.2 High Pass Filter

A high-pass filter blocks all frequencies from 0 up to f_c and passes all frequencies above f_c . It is the complement (mirror image) of the LPF.

Fig 21-2: Ideal High-Pass Filter Response — Stopband (0 to f_c) then Passband

- Stopband: 0 to f_c — low frequencies are blocked
- Passband: above f_c — high frequencies pass unattenuated

- Ideal HPF: zero attenuation in passband, infinite in stopband, vertical transition

9.3 Bandpass Filter

A bandpass filter passes only the band of frequencies between f_1 and f_2 , blocking all outside. Key equations:

$$BW = f_2 - f_1 \quad | \quad f_0 = \sqrt{f_1 \times f_2} \quad | \quad Q = f_0 / BW$$

Fig 21-3: Ideal Bandpass Filter Response — Passband between f_1 and f_2 , BW shown

- $Q < 1 \rightarrow$ Wideband filter. $Q > 1 \rightarrow$ Narrowband filter.
- $f_0 =$ geometric mean $= \sqrt{f_1 \times f_2}$. $Q =$ center frequency / bandwidth.

9.4 Bandstop Filter

A bandstop (notch) filter passes all frequencies EXCEPT the band f_1 to f_2 . It 'notches out' unwanted frequencies. Same BW, f_0 , Q definitions as bandpass.

Fig 21-4: Ideal Bandstop Filter Response — passes below f_1 and above f_2 , blocks f_1 to f_2

Bandstop Application

Remove 50Hz/60Hz power-line hum from audio or ECG signals. The filter must have high Q to only remove the hum frequency without affecting nearby frequencies.

9.5 All-Pass Filter

The all-pass filter passes ALL frequencies with equal amplitude (unity gain) but introduces frequency-dependent phase shift. Used for phase equalization and time delay.

Fig 21-5: Ideal All-Pass Filter Response — flat gain at all frequencies, phase varies

10. First Order Filter Stages

First-order (1-pole) active filter stages have only ONE capacitor $\rightarrow$ can produce only LPF or HPF response. Roll-off rate = -20 dB/decade. Bandpass and bandstop need $n > 1$ poles.

10.1 First Order Low Pass Stages

Figure 21-21 below shows all three first-order LPF variants. From top to bottom: (a) unity-gain noninverting, (b) noninverting with gain, (c) inverting with gain.

Fig 21-21: First-Order Low-Pass Stages — (a) Unity gain; (b) Non-inverting with gain; (c) Inverting with gain

(a) Unity-Gain Non-Inverting LPF

$$A_v = 1 \quad | \quad f_c = 1 / (2\pi \times R_1 \times C_1)$$

Simplest LPF — RC lag circuit + voltage follower. $A_v = 1$. At DC, capacitor is open, full input reaches op-amp. Above f_c , capacitive reactance decreases → output rolls off.

(b) Non-Inverting LPF with Voltage Gain

$$A_v = R_2/R_1 + 1 \quad | \quad f_c = 1 / (2\pi \times R_3 \times C_1) \quad \dots \quad (21-10, 21-11)$$

R_1 and R_2 set the midband gain independently from f_c (set by R_3C_1). Above f_c , R_3C_1 lag circuit reduces the noninverting input voltage → output rolls off at -20 dB/decade.

(c) Inverting LPF with Gain

$$A_v = -R_2/R_1 \quad | \quad f_c = 1 / (2\pi \times R_2 \times C_1) \quad \dots \quad (21-12, 21-13)$$

C_1 in series with R_2 in feedback branch. At low f , C_1 acts open → gain = $-R_2/R_1$. As f increases, X_{C_1} decreases → feedback impedance drops → gain falls to zero. $f_c = 1/(2\pi R_2 C_1)$.

10.2 First Order High Pass Stages

Figure 21-22 below shows all three first-order HPF variants. R and C positions are swapped compared to LPF: capacitor now in series with signal path, resistor to ground.

Fig 21-22: First-Order High-Pass Stages — (a) Unity gain; (b) Non-inverting with gain; (c) Inverting with gain

(a) Unity-Gain Non-Inverting HPF

$$A_v = 1 \quad | \quad f_c = 1 / (2\pi \times R_1 \times C_1) \quad \dots \quad (21-14)$$

At DC ($f=0$), C_1 is open → zero signal reaches op-amp → output = 0. As f increases, X_{C_1} decreases → more signal passes → above f_c , output $\approx$ input ($A_v = 1$).

(b) Non-Inverting HPF with Gain

$$A_v = R_2/R_1 + 1 \quad | \quad f_c = 1 / (2\pi \times R_3 \times C_1) \quad \dots \quad (21-15, 21-16)$$

At high frequencies: gain = $R_2/R_1 + 1$. Below f_c , R_3C_1 blocks signal, output rolls off at -20 dB/decade.

(c) Inverting HPF

$$A_v = -C_1/C_2 \quad | \quad f_c = 1 / (2\pi \times R_1 \times C_2) \quad \dots \quad (21-17, 21-18)$$

At high f , both capacitors are shorts → gain = $-X_{C_2}/X_{C_1} = -C_1/C_2$. At low f , large X_C → signal blocked.

11. VCVS Unity-Gain Second-Order Low Pass Filters

Second-order (2-pole) filters roll off at -40 dB/decade — twice the rate of first-order. These Sallen-Key (VCVS) filters use the op-amp as a unity-gain voltage follower. Features: two equal resistors (R), two different capacitors (C_1 and C_2), C_2 provides positive feedback path for sharper roll-off.

11.1 Sallen-Key Low Pass Filter — Butterworth & Bessel

Figure 21-24 below shows the Sallen-Key second-order LPF for Butterworth and Bessel responses. Note: two equal R values, two different C values, op-amp as voltage follower ($A_v = 1$):

Fig 21-24: Second-Order VCVS (Sallen-Key) Low-Pass Stage — Butterworth ($Q=0.707$, $K_c=1$) and Bessel ($Q=0.577$, $K_c=0.786$)

$A_v = 1$		$Q = 0.5 \times \sqrt{(C_2/C_1)} / (2\pi R \sqrt{C_1 C_2})$		$f_p = 1 / (2\pi R \sqrt{C_1 C_2})$
$f_c = K_c \times f_p$	(Butterworth: $K_c=1$ Bessel: $K_c=0.786$) ... (21-21)			

Response	Q Value	K_c	Characteristic
Butterworth	$Q = 0.707$	$K_c = 1$	Flattest passband, $f_c = f_p$
Bessel	$Q = 0.577$	$K_c = 0.786$	Linear phase, $f_c = 0.786 \times f_p$

Example 21-3 — Butterworth (from textbook Fig 21-27):

Example 21-3: Butterworth Unity-Gain Example — $R=30k\Omega$, $C_1=820pF$, $C_2=1.64nF \rightarrow Q=0.707 \rightarrow f_c=f_p=4.58kHz$

Example 21-4 — Bessel (from textbook Fig 21-28):

Example 21-4: Bessel Unity-Gain Example — $R=51k\Omega$, $C_1=330pF$, $C_2=440pF \rightarrow Q=0.577 \rightarrow f_c=0.786 \times 8.19kHz=6.44kHz$

12. VCVS Equal-Component Low Pass Filter

Figure 21-31 (below) shows the equal-component Sallen-Key LPF. Both resistors AND both capacitors have the same values ($R_1=R_2=R$, $C_1=C_2=C$). The op-amp provides gain > 1 via R_1 , R_2 feedback network. Q becomes a function of voltage gain — must keep $A_v < 3$ to avoid oscillation!

Fig 21-31: VCVS Equal-Component Stage — same R and C values, gain $A_v = R_2/R_1 + 1$, $Q = 1/(3-A_v)$

$A_v = R_2/R_1 + 1$	... (21-29)
$Q = 1 / (3 - A_v)$	... (21-30)  If $A_v \geq 3 \rightarrow$ OSCILLATES!
$f_p = 1 / (2\pi \times R \times C)$	(since $R_1=R_2=R$, $C_1=C_2=C$)

- Butterworth ($Q=0.707$): Solve $\rightarrow A_v = 3 - 1/0.707 = 1.586$
- Bessel ($Q=0.577$): Solve $\rightarrow A_v = 3 - 1/0.577 = 1.268$
- CAUTION: A_v must be < 3 — otherwise $Q \rightarrow \infty$ and circuit self-oscillates!

Equal-Component vs Unity-Gain

Unity-Gain Sallen-Key: Q set by C_2/C_1 ratio. Two different C values. One R value. Equal-

Component: Q set by gain (R_2/R_1 ratio). One C value. One R value. But gain MUST be < 3 .

13. VCVS High Pass Filter

The VCVS HPF (Fig 21-35) is obtained from the Sallen-Key LPF by swapping R and C positions. Capacitors go in series with the signal path; resistors connect to ground. The analysis is identical except $f_c = f_p / K_c$ (divide, not multiply):

$$f_c = f_p / K_c \quad \dots (21-31) \quad (\text{HPF: DIVIDE by } K_c - \text{opposite of LPF!})$$

Fig 21-35: Second-Order VCVS High-Pass Stages — (a) Unity gain with $Q=0.5\sqrt{(R_1/R_2)}$; (b) Equal-component with $Q=1/(3-A_v)$

Unity-Gain VCVS HPF (Fig 21-35a)

$$A_v = 1 \quad | \quad Q = 0.5 \times \sqrt{(R_1/R_2)} \quad | \quad f_p = \frac{1}{2\pi C \sqrt{(R_1 R_2)}}$$

Key difference from LPF: Q depends on RESISTOR ratio (R_1/R_2), not capacitor ratio.

Equal-Component VCVS HPF (Fig 21-35b)

$$A_v = R_2/R_1 + 1 \quad | \quad Q = 1/(3 - A_v) \quad | \quad f_p = \frac{1}{2\pi RC}$$

Example 21-9 — High-Pass Butterworth (from textbook Fig 21-36):

Example 21-9 (Fig 21-36): High-Pass Butterworth — $C=4.7\text{nF}$, $R_1=24\text{k}\Omega$, $R_2=12\text{k}\Omega \rightarrow Q=0.707 \rightarrow f_c=f_p=2\text{kHz}$, rolls off at -40dB/dec below 2kHz

14. MFB (Multiple Feedback) Bandpass Filter

The MFB bandpass filter uses an inverting op-amp configuration with TWO feedback paths — one through capacitor C and one through resistor R_2 . At low frequencies, capacitors are open $\rightarrow$ no signal $\rightarrow$ output = 0. At high frequencies, capacitors are shorted $\rightarrow$ gain = 0. Between extremes, circuit acts as inverting bandpass amplifier.

$$BW = f_2 - f_1 \quad | \quad f_0 = \sqrt{(f_1 \times f_2)} \quad | \quad Q = f_0 / BW$$

14.1 Wideband BPF ($Q < 1$) — Fig 21-38

When $Q < 1$: use cascaded HPF + LPF stages (wideband approach). High-pass sets f_1 ; low-pass sets f_2 . Only band between f_1 and f_2 passes both stages.

Fig 21-38: Wideband BPF — High-pass ($f_c=300\text{Hz}$) cascaded with Low-pass ($f_c=3.3\text{kHz}$). Combined response shown below.

Wideband Example (from notes)

$f_1=300\text{Hz}$, $f_2=3.3\text{kHz}$ → $f_0=\sqrt{(300 \times 3300)}=995\text{Hz}$, $\text{BW}=3\text{kHz}$, $Q=0.332 < 1$ → Wideband → use cascaded HPF+LPF.

14.2 Narrowband MFB Filter ($Q > 1$) — Fig 21-39

When $Q > 1$: use the MFB (Multiple Feedback) circuit shown in Figure 21-39 below:

Fig 21-39: MFB Narrowband Bandpass Stage — two equal capacitors C , resistors R_1 and R_2 . Input to inverting terminal.

$A_v = -R_2 / (2R_1)$... (21-32)	(Gain at center frequency f_0)
$Q = 0.5 \times \sqrt{(R_2/R_1)}$... (21-33)	
$Q = 0.707 \times \sqrt{(-A_v)}$... (21-34)	(Equivalent expression)
$f_0 = 1 / (2\pi \times C \times \sqrt{(R_1 \times R_2)})$	(Both capacitors = C)

MFB Bandpass Filter Summary	
Gain at f_0	$A_v = -R_2/(2R_1)$ [inverting, 180° phase]
Q factor	$Q = 0.5 \times \sqrt{(R_2/R_1)} = 0.707 \times \sqrt{ A_v }$
Center Frequency	$f_0 = 1/(2\pi C \sqrt{(R_1 R_2)})$
For $Q < 1$	Use cascaded LPF + HPF (wideband)
For $Q > 1$	Use MFB narrowband circuit (Fig 21-39)
Input	Goes to INVERTING terminal

15. Bandstop Filters

Bandstop (notch) filters reject a specific band f_1 to f_2 while passing all other frequencies. The stopband is the rejected band. Same BW, f_0 , Q definitions as bandpass. Sometimes called notch filter — it 'notches out' specific frequencies.

- Stopband: f_1 to f_2 — infinite attenuation (ideal)
- Passband: below f_1 AND above f_2 — zero attenuation (ideal)
- Two vertical transitions: at f_1 and at f_2 (ideal case)
- $Q \ll 1$ → wide notch. $Q \gg 1$ → sharp narrow notch.

Implementation Methods

- Twin-T passive network + op-amp feedback: Classic narrow notch filter. The Twin-T RC network has a natural null at $f_0 = 1/(2\pi RC)$. Adding op-amp feedback sharpens the notch.
- State-variable filter: Simultaneously provides LP, HP, and BP outputs. Summing LP + HP outputs gives bandstop response.

- Wideband approach ($Q < 1$): Sum LPF output + HPF output → signals in passband add constructively; signals in stopband cancel.

Classic Application: 50Hz/60Hz Notch Filter

A notch filter tuned to 50Hz (India/Europe) or 60Hz (USA) removes power-line hum from audio recordings, ECG/EKG medical signals, and sensitive instrumentation. High Q ensures only the hum frequency is removed without affecting 40Hz bass or 80Hz tones.

⚡ QUICK REVISION GUIDE — Module 4

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⚡ NEGATIVE FEEDBACK — Four Types Quick Reference

- VCVS (Voltage Amp): $z_{in} = \infty$, $z_{out} = 0$ | $A_v = v_{out}/v_{in} \approx R_f/R_1 + 1 = 1/B$
- ICVS (Transresistance Amp): $z_{in} = 0$, $z_{out} = 0$ | $r_m = v_{out}/i_{in} = -R_f$ (Ohms)
- VCIS (Transconductance Amp): $z_{in} = \infty$, $z_{out} = \infty$ | $g_m = i_{out}/v_{in} = 1/R_1$ (Siemens)
- ICIS (Current Amp): $z_{in} = 0$, $z_{out} = \infty$ | $A_i = i_{out}/i_{in} \approx R_2/R_1 + 1$

⚡ VCVS KEY FORMULAS (Equations 19-1 to 19-12)

- $B = R_1/(R_1 + R_f)$ [feedback fraction — always between 0 and 1]
- Exact $A_v = A_{VOL} / (1 + A_{VOL} \times B)$
- Ideal $A_v \approx 1/B = R_f/R_1 + 1$ [when $A_{VOL} \times B \gg 1$]
- % Max Error = $100\% / (1 + A_{VOL}(\min) \times B)$
- $z_{in}(CL) \approx (1 + A_{VOL} \times B) \times R_{in} \rightarrow$ INCREASES with NFB
- $z_{out}(CL) = R_{out} / (1 + A_{VOL} \times B) \rightarrow$ DECREASES with NFB
- $THD_{CL} = THD_{OL} / (1 + A_{VOL} \times B) \rightarrow$ DECREASES with NFB

⚡ ACTIVE FILTERS — Cutoff Frequency Master List

- 1st Order LPF (unity): $A_v=1$, $f_c = 1/(2\pi R_1 C_1)$
- 1st Order LPF (with gain): $A_v = R_2/R_1 + 1$, $f_c = 1/(2\pi R_3 C_1)$
- 1st Order LPF (inverting): $A_v = -R_2/R_1$, $f_c = 1/(2\pi R_2 C_1)$
- 1st Order HPF (unity): $A_v=1$, $f_c = 1/(2\pi R_1 C_1)$ [C and R swapped vs LPF]
- Sallen-Key 2nd LPF: $Q=0.5\sqrt{C_2/C_1}$, $f_p=1/(2\pi R\sqrt{C_1 C_2})$, $f_c=K_c \times f_p$
- Equal-Component LPF/HPF: $Q=1/(3-A_v)$, $f_p=1/(2\pi RC)$, $A_v < 3$!!
- VCVS HPF unity: $Q=0.5\sqrt{R_1/R_2}$, $f_p=1/(2\pi C\sqrt{R_1 R_2})$, $f_c = f_p/K_c$
- MFB Bandpass: $A_v=-R_2/(2R_1)$, $Q=0.5\sqrt{R_2/R_1}$, $f_0=1/(2\pi C\sqrt{R_1 R_2})$

⚡ BUTTERWORTH vs BESSEL vs FILTER ORDER

- Butterworth: $Q=0.707$, $K_c=1 \rightarrow f_c = f_p$. FLATTEST passband. No ripple.

- Bessel: $Q=0.577$, $K_c=0.786 \rightarrow f_c = 0.786 \times f_p$. LINEAR PHASE. Preserves pulse shapes.
- LPF $f_c = K_c \times f_p$ | HPF $f_c = f_p / K_c$ [DIVIDE for HPF!]
- 1st order: -20dB/decade | 2nd order: -40dB/decade | nth order: -20n dB/decade

⚡ MUST-REMEMBER VTU EXAM POINTS

- NFB ALWAYS reduces gain but improves ALL: bandwidth, distortion, impedance stability.
- VCVS: $z_{in} \uparrow (\times(1+A_{VOL} \times B))$ AND $z_{out} \downarrow (\div(1+A_{VOL} \times B))$.
- ICVS: BOTH z_{in} and $z_{out} \rightarrow$ ZERO ($\div$ by A_{VOL} , not $A_{VOL} \times B$).
- VCIS: BOTH z_{in} and $z_{out} \rightarrow$ INFINITY ($\times$ by A_{VOL} or $A_{VOL} \times B$).
- ICIS: $z_{in} \rightarrow$ ZERO, $z_{out} \rightarrow$ INFINITY.
- Ideal gain for VCVS non-inv: $A_v = 1/B = 1 + R_f/R_1$ (does NOT depend on A_{VOL}).
- Active filters use op-amps + RC — NO inductors needed.
- Equal-component: $Q=1/(3-A_v)$. DANGER: if $A_v \geq 3 \rightarrow$ circuit OSCILLATES!
- For HPF: $f_c = f_p/K_c$ (DIVIDE). For LPF: $f_c = K_c \times f_p$ (MULTIPLY). Don't mix these up!
- MFB Bandpass input goes to INVERTING terminal — output is phase-inverted (180°).
- $Q < 1 \rightarrow$ Wideband $\rightarrow$ cascaded HPF + LPF. $Q > 1 \rightarrow$ Narrowband $\rightarrow$ MFB circuit.
- $THD_{CL} = THD_{OL}/(1+A_{VOL} \times B)$. NFB reduces distortion by loop gain factor.
- Sallen-Key: Two equal R values, two DIFFERENT C values (or vice versa for equal-comp).

All the best! — vtu-easy.com

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